

# UTFT\_Geometry

Add-on Library for UTFT: Geometric functions

## Manual



## Introduction:

This library is an add-on to UTFT and will not work on its own.

This library adds some geometric functions to UTFT which are not included in the main library.

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You can always find the latest version of the library at <http://www.RinkyDinkElectronics.com/>

For version information, please refer to **version.txt**.

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## FUNCTIONS:

| UTFT_Geometry(UTFT);        |                                                                                                                                                                                                                    |
|-----------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| The main class constructor. |                                                                                                                                                                                                                    |
| Parameters:                 | UTFT: A reference to an already created UTFT object ( <i>remember the '&amp;' in front of the object name</i> )                                                                                                    |
| Usage:                      | UTFT_Geometry geo(&myGLCD); // Start an instance of UTFT_Geometry pointing to the UTFT object myGLCD                                                                                                               |
| Notes:                      | Remember that all functions in UTFT_Geometry must be called through the UTFT_Geometry object (geo in this example), while all UTFT functions still must be called through the UTFT object (myGLCD in this example) |

| drawTriangle(x1, y1, x2, y2, x3, y3);       |                                                                                                                                                                                                                                                |
|---------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Draw a triangle with the specified corners. |                                                                                                                                                                                                                                                |
| Parameters:                                 | x1: x-coordinate of the first corner<br>y1: y-coordinate of the first corner<br>x2: x-coordinate of the second corner<br>y2: y-coordinate of the second corner<br>x3: x-coordinate of the third corner<br>y3: y-coordinate of the third corner |
| Usage:                                      | geo. drawTriangle(159,119,319,119,319,239); // Draw a triangle                                                                                                                                                                                 |

| fillTriangle(x1, y1, x2, y2, x3, y3);              |                                                                                                                                                                                                                                                |
|----------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Draw a filled triangle with the specified corners. |                                                                                                                                                                                                                                                |
| Parameters:                                        | x1: x-coordinate of the first corner<br>y1: y-coordinate of the first corner<br>x2: x-coordinate of the second corner<br>y2: y-coordinate of the second corner<br>x3: x-coordinate of the third corner<br>y3: y-coordinate of the third corner |
| Usage:                                             | geo. fillTriangle(159,119,319,119,319,239); // Draw a filled triangle                                                                                                                                                                          |

| drawArc(x, y, r, startAngle, endAngle[, thickness]); |                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Draw an arc.                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| Parameters:                                          | x : x-coordinate for the center of the imaginary circle<br>y : y-coordinate for the center of the imaginary circle<br>r : Radius of the imaginary circle<br>startAngle: Angle from the center of the imaginary circle to start drawing the arc<br>endAngle : Angle from the center of the imaginary circle to stop drawing the arc<br>thickness : <b>&lt;optional&gt;</b><br>Thickness of the arc in pixels. Default is 1 pixel. |
| Usage:                                               | geo. drawArc(159,239,100,-75,75); // Draw an arc                                                                                                                                                                                                                                                                                                                                                                                 |
| Notes:                                               | 0° is at the top of the imaginary circle                                                                                                                                                                                                                                                                                                                                                                                         |

| drawPie(x, y, r, startAngle, endAngle);                       |                                                                                                                                                                                                                                                                                                                                    |
|---------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Draw an arc with lines to the center of the imaginary circle. |                                                                                                                                                                                                                                                                                                                                    |
| Parameters:                                                   | x : x-coordinate for the center of the imaginary circle<br>y : y-coordinate for the center of the imaginary circle<br>r : Radius of the imaginary circle<br>startAngle: Angle from the center of the imaginary circle to start drawing the arc<br>endAngle : Angle from the center of the imaginary circle to stop drawing the arc |
| Usage:                                                        | geo. drawPie(159,50,100,135,225); // Draw a pie shape                                                                                                                                                                                                                                                                              |
| Notes:                                                        | 0° is at the top of the imaginary circle                                                                                                                                                                                                                                                                                           |